CPTG Geometric Pi Branch: Comparison Coordinates

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Abstract

Curvature Polarization Transport Gravity (CPTG) provides a geometric cosmological branch that can be expressed directly in the observational comparison coordinates used by modern cosmological surveys. The geometric pi branch supplies a fixed native structure for expansion, acoustic transport, clock behavior, baryon conversion, growth, lensing, and distance-ruler comparisons. Once the native branch is fixed, its quantities can be translated into the coordinate systems used by CMB likelihoods, supernova distance-redshift tests, BBN abundance comparisons, weak-lensing S_8 summaries, BAO measurements, ShapeFit vectors, and full-shape diagnostic pipelines.

This article gives a standalone map of those comparison coordinates. It lists the locked CPTG branch values, the sector-by-sector translation rules, the effective comparison rows, and the diagnostic results obtained when the same fixed CPTG construction is carried across independent observational sectors. The purpose is to show how one geometric branch can generate coherent comparison quantities across expansion, clock, baryon, acoustic, growth, lensing, and ruler channels while keeping the CPTG-native theory structure explicit.

1 Introduction

Curvature Polarization Transport Gravity (CPTG) [17, 18] provides a geometric framework whose cosmological branch can be carried into the observational coordinates used by modern survey pipelines. The geometric pi branch is defined inside CPTG first, then expressed in the comparison languages used by supernova, CMB, BBN, weak-lensing, BAO, ShapeFit, and large-scale-structure analyses. This gives the theory a practical bridge between native curvature geometry and the quantities reported by observational programs.

The strength of this structure is that one fixed geometric branch can generate multiple comparison-layer predictions. The same branch that defines the expansion response can be translated into distance-redshift behavior for supernovae, clock and lookback-time behavior for age tests, baryon and expansion quantities for BBN, acoustic comparison quantities for the CMB, compressed growth and lensing quantities for S_8, and distance-ratio coordinates for BAO and ShapeFit tests. The result is a unified comparison framework rather than a separate construction for each observational sector.

In this sense, CPTG is capable of connecting background expansion, acoustic transport, baryon conversion, clock normalization, growth, lensing, and distance-ruler behavior through a single geometric organizing principle. Its comparison coordinates allow the same underlying branch to be tested against independent observational summaries without changing the native geometric construction. Each sector becomes a different projection of the same curvature-transport structure.

The geometric pi branch is especially useful because it provides a compact expansion law with fixed transport weights. Once the branch is locked, the observational work becomes a question of coordinate expression: how the CPTG-native quantities appear when written in the variables used by supernova Hubble diagrams, CMB likelihoods, BBN abundance summaries, weak-lensing S_8 measurements, DESI BAO ratios, and ShapeFit coordinate vectors. This makes CPTG scalable across observational domains while preserving a common theoretical core.

This article describes that comparison-coordinate capability. It shows how the locked CPTG geometric branch can be expressed in the observational languages needed to compare with the major cosmological data sectors. The emphasis is on how a single geometric theory can produce coherent, sector-by-sector comparison quantities across the expansion, clock, baryon, acoustic, growth, lensing, and ruler channels.

2 Purpose and Scope

CPTG comparison work begins from fixed native quantities and then maps those quantities into observational coordinates. This order matters. The comparison coordinate is not the theory. It is the language required by a particular data product, likelihood, or compressed observational summary.

The comparison hierarchy is

$$\begin{gathered}\text{CPTG-native geometric quantity}\\ \downarrow\\ \text{comparison-coordinate quantity}\\ \downarrow\\ \text{reported observational summary}\end{gathered} \tag{1}$$

The first layer is fixed by the geometric branch. The second layer expresses the native result in the variables used by a particular observational pipeline. The third layer is the published or reconstructed quantity against which the comparison is made.

The branch values used here are

$$p_C = \pi, \tag{2}$$

$$p_{\rm ac} = 3 - \frac{\pi}{100} = 2.968584073464, \tag{3}$$

$$\mathcal{G}_T = \frac{p_{\rm ac}}{p_C} = 0.9449296585513721, \tag{4}$$

and

$$\sqrt{\mathcal{G}_T} = 0.9720749243506758. \tag{5}$$

The native geometric acceleration anchor is

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} = 5.568327996831708 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{6}$$

The raw geometric normalization is

$$H_0^{(\pi)} = 69.4162507897\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{7}$$

and the acoustic/CMB comparison normalization is

$$H_{0,\mathrm{CMB}}^{\mathrm{CPTG}} = 67.4777967351\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{8}$$

The framework and public computational materials are documented in Refs. [17, 18].

3 Parameter Inventory

This parameter inventory collects the fixed CPTG-native branch values and comparison-coordinate symbols used throughout the article. These definitions provide the common reference layer for the CMB, supernova, BBN, weak-lensing, BAO, ShapeFit, and full-shape diagnostic sections that follow.

Quantity	Meaning or definition	Locked value	Main location
p_C	Native curvature exponent	π	Use / branch definition
p_{ac}	Acoustic exponent	$3 - \pi/100 = 2.968584073464$	Use: acoustic layer
$\mathcal{G}_T$	Acoustic transport factor	$p_{\mathrm{ac}}/p_C = 0.9449296585513721$	Use: CMB gates
$\sqrt{\mathcal{G}_T}$	Hubble-rate transport factor	0.9720749243506758	Use: Hubble/CMB projection
$\mathcal{G}_T^2$	Two-step visibility / optical-depth transport gate	0.8928920596100127	Visibility gate; CMB closure
$\mathcal{B}_\pi$	Late/geometric branch weight	$1 - 1/\pi = 0.681690113816$	Distance layer
$\mathcal{T}_\pi$	Transport branch weight	$1/\pi = 0.318309886184$	Distance layer
$a_{\pi,0}$	Native geometric acceleration anchor	$5.568327996831708 \times 10^{-10}\ \mathrm{m\,s^{-2}}$	Use: geometric branch
H_∞	Native asymptotic Hubble scale	$57.3131992073\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Distance layer
$H_0^{(\pi)}$	Raw geometric Hubble normalization	$69.4162507897\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Use: Hubble comparison
$H_{0,\mathrm{CMB}}^{\mathrm{CPTG}}$	Acoustic/CMB comparison Hubble normalization	$67.4777967351\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Use: CMB row
$H_{\infty,\mathrm{CMB}}^{\mathrm{CPTG}}$	Transported asymptotic comparison scale	$55.7127237837\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Distance layer
R_C	Native branch curvature scale	788.7823091477 Gpc	Distance layer
$t_{0,\mathrm{clock}}^{\mathrm{CPTG}}$	Clock-branch age	13.88517358 Gyr	Clock branch / age test

Table 1: Native branch, transport, and clock parameters.

Quantity	Meaning or definition	Locked value	Main location
$\Omega_{m,\mathrm{ac}}$	Acoustic matter comparison fraction	0.314976552918	Distance layer
$\Omega_{\Lambda,\mathrm{ac}}$	Complementary acoustic branch fraction	0.685023447082	Distance layer
$E_\pi(a)^2$	Native pi-branch expansion law	$\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}$	Distance layer
$E_\pi(z)^2$	Redshift-form pi-branch expansion law	$\mathcal{B}_\pi + \mathcal{T}_\pi (1+z)^\pi$	Distance layer
dt_{geom}	Native geometric clock element	$da/[aH_{\mathrm{geom}}(a)]$	Clock branch
$\Xi_i(a)$	Branch-specific observed-time projection	channel dependent	Clock branch
dt_i	Observed comparison-time element	$\Xi_i(a) dt_{\mathrm{geom}}$	Clock branch
$t_i(a)$	Branch comparison age integral	$\int_0^a \Xi_i(a')\, da'/[a' H_{\mathrm{geom}}(a')]$	Clock branch
$\mathcal{L}_{\mathrm{loc}}$	Local luminosity-distance comparison projection	1.052203182527	Hubble comparison

Table 2: Expansion, distance, and time-coordinate parameters.

Quantity	Meaning or definition	Locked value	Main location
$\mathcal{D}_c$	Cold-sector comparison transport	0.9735353159758136	CMB acoustic transport
$\mathcal{D}_c$	Geometric form	$\sqrt{\mathcal{G}_T}\,(1 + (\pi-3)/(30\pi))$	CMB acoustic transport
$\omega_{c,\mathrm{CPTG}} h^2$	Native cold-sector density before comparison transport	0.120885025627	CMB gate closure
$\omega_{c,\mathrm{eff}} h^2$	Effective CMB cold-sector comparison density	0.117685841620526	CMB row
$\omega_b h^2$	Effective CMB baryon density	0.022527857494	CMB row
A_s^{CPTG}	Native scalar-amplitude normalization	2.04×10^{-9}	Amplitude gate
$\mathcal{A}_s$	Scalar-amplitude comparison gate	$\pi/3 = 1.047197551197$	Amplitude gate
A_s^{eff}	Effective CMB scalar amplitude	$2.136283004441 \times 10^{-9}$	CMB row
τ_{CPTG}	Native optical-depth normalization	0.066	Visibility gate
$\mathcal{T}_\tau$	Optical-depth comparison gate	$\mathcal{G}_T^2 = 0.8928920596100127$	Visibility gate
τ_{eff}	Effective CMB optical-depth comparison value	0.05893087593426084	CMB row
n_s	CMB scalar spectral index comparison value	0.968584073464	CMB row
N_{eff}	CMB radiation comparison value	2.968584073464	CMB row; BBN
Y_{He}	CMB helium comparison value	0.2485	CMB row
A_{lens}	CMB lensing-amplitude setting	1	CMB row

Table 3: Locked CMB comparison-layer parameters.

Quantity	Meaning or definition	Locked value	Main location
f_b	Baryon fraction	$\pi/20 = 0.157079632679$	Baryon conversion
$\Omega_b h^2_{\rm ac}$	Acoustic/CMB baryon-density conversion	0.0225278574944	Baryon conversion
$\eta_{10}^{\rm ac}$	Acoustic/CMB baryon-to-photon conversion	6.170380167710	Baryon conversion
$\Omega_b h^2_{\rm BBN}$	Transport-scaled BBN baryon conversion	0.021898765370	BBN comparison
$\eta_{10}^{\rm BBN}$	Transport-scaled BBN baryon-to-photon conversion	5.998071834744	BBN comparison
$N_{\rm eff}^{\rm CPTG}$	Effective-radiation comparison value	2.968584073464	BBN comparison
$Y_p^{\rm network}$	Network helium prediction used in comparison	0.245694338195	BBN comparison
$10^6({\rm D/H})_{\rm network}$	Network deuterium prediction used in comparison	25.074383114428	BBN comparison

Table 4: Baryon, radiation, and light-element comparison parameters.

Quantity	Meaning or definition	Locked value	Main location
$A_L^{\rm CPTG}$	Native lensing response scale	$(p_C/p_{\rm ac})^2 = 1.119956202138$	Growth/lensing
$\mu_{\rm CPTG}(a)$	Reduced growth response	$1 + (\pi/2)(A_L^{\rm CPTG} - 1)(1 - a^{p_{\rm ac}})$	Growth/lensing
$\Sigma_{\rm CPTG}(a)$	Reduced lensing response	$1 + (A_L^{\rm CPTG} - 1)(1 - a^{p_{\rm ac}})$	Growth/lensing
α_C	Late compression index	$1 - 2/\pi$	Growth/lensing
$q_{\rm bandpower}$	CMB-lensing bandpower compression factor	$(1 - \mathcal{G}_T)/\mathcal{B}_\pi$	Growth/lensing
$A_{\rm bandpower}$	Raw CMB-lensing bandpower conversion	1.009690662776	Growth/lensing
$S_{8,\rm late}^{\rm CPTG}$	Late-time weak-lensing comparison amplitude	0.788071335232	Weak-lensing comparison
$S_{8,\rm CMBL}^{\rm CPTG}$	Compressed CMB-lensing amplitude	0.821191065949	CMB-lensing comparison

Table 5: Growth and lensing comparison parameters.

Quantity	Meaning or definition	Locked value	Main location
$q_\perp$	Transverse AP comparison coordinate	$D_M^{\rm CPTG}/D_M^{\rm fid}$, w/ ruler conversion when used	DESI ShapeFit / BAO
$q_\parallel$	Radial AP comparison coordinate	$D_H^{\rm CPTG}/D_H^{\rm fid}$, w/ ruler conversion when used	DESI ShapeFit / BAO
$q_{\rm iso}$	Isotropic AP compression	$(q_\perp^2 q_\parallel)^{1/3}$	DESI ShapeFit
$q_{\rm ap}$	AP anisotropy coordinate	$q_\parallel/q_\perp$	DESI ShapeFit
dm	ShapeFit spectral-shape coordinate	0 unless explicitly derived	DESI ShapeFit
df	ShapeFit growth-rate coordinate	$f_{\rm CPTG}/f_{\rm fid}$	DESI ShapeFit
$\chi^2_{\rm identity}$	Native identity check	0	DESI ShapeFit
$\chi^2_{\rm no\ shift}$	No-shift baseline	38.70438441	DESI ShapeFit
$\chi^2_{\rm CPTG}$	DESI-like fiducial compressed ShapeFit result	35.19040980558847	DESI ShapeFit
χ^2_ν	Reduced compressed ShapeFit result over 24 coordinates	1.466267075232853	DESI ShapeFit
$\Delta\chi^2$	CPTG shift relative to no-shift baseline	-3.51397460441153	DESI ShapeFit
$r_d^{\rm ratio}$	BAO ruler-ratio coordinate	$\mathcal{G}_T^{-1/4} = 1.014261942163$	BAO quarter ruler
$\chi^2_{\rm BAO}(q=1)$	BAO unity-wrapper baseline	20.12426210852749	BAO wrapper
$\chi^2_{\rm BAO}(\rm best)$	Best scanned BAO ruler coordinate	13.501376842627	BAO wrapper
$\chi^2_{\rm BAO}(\mathcal{G}_T^{-1/4})$	Quarter-ruler BAO coordinate result	13.505973446003	BAO wrapper

Table 6: DESI compressed-coordinate and BAO wrapper parameters.

Article location	Parameter families located there	Purpose
Purpose and Scope	p_C, $p_{\rm ac}$, $\mathcal{G}_T$, $\sqrt{\mathcal{G}_T}$, $a_{\pi,0}$, $H_0^{(\pi)}$, $H_{0,\rm CMB}^{\rm CPTG}$	Establishes the fixed branch values.
Distance and Acoustic-Ruler Comparison Layer	$\mathcal{B}_\pi$, $\mathcal{T}_\pi$, $E_\pi(a)$, $E_\pi(z)$, H_∞, R_C, $\Omega_{m,\rm ac}$, $\Omega_{\Lambda,\rm ac}$, $r_d^{\rm ratio}$	Gives distance, expansion, and ruler coordinates.
Baryon and Light-Element Conversion	f_b, $\Omega_b h^2_{\rm ac}$, $\eta_{10}^{\rm ac}$, $\Omega_b h^2_{\rm BBN}$, $\eta_{10}^{\rm BBN}$, $N_{\rm eff}^{\rm CPTG}$, Y_p, D/H	Gives baryon and BBN conversion coordinates.
Perturbation, Growth, and Lensing Channels	$A_L^{\rm CPTG}$, $\mu_{\rm CPTG}$, $\Sigma_{\rm CPTG}$, $A_{\rm bandpower}$, $S_{8,\rm late}^{\rm CPTG}$, $S_{8,\rm CMBL}^{\rm CPTG}$	Gives growth and lensing compression coordinates.
Locked CMB Acoustic Transport Layer	$\mathcal{D}_c$, $\omega_{c,\rm CPTG}h^2$, $\omega_{c,\rm eff}h^2$, $A_s^{\rm CPTG}$, $A_s^{\rm eff}$, $\tau_{\rm CPTG}$, $\tau_{\rm eff}$, $A_{\rm lens}$	Gives the locked CMB comparison row and closure maps.
Geometric Origin of the CMB Amplitude and Visibility Gates	$\mathcal{A}_s = \pi/3$, $\mathcal{T}_\tau = \mathcal{G}_T^2$, $\mathcal{D}_c(p_{\rm ac})$, $\tau_{\rm eff}(p_{\rm ac})$	Gives the gate origin and perturbation checks.
Compressed Validation Suite Results	Pantheon+, age/clock, BBN, weak-lensing, and CMB-lensing comparison outputs	Gives sector diagnostic results.
DESI DR1 Compressed-Coordinate, BAO Quarter-Ruler, and Full-Shape Diagnostics	$q_\perp$, $q_\parallel$, $q_{\rm iso}$, $q_{\rm ap}$, dm, df, $r_d^{\rm ratio}$, ShapeFit and BAO χ^2 values	Gives DESI and BAO coordinate tests.
Validation Status and Falsifiability	Claim-level labels, pass/fail criteria, and forbidden overclaims	Defines what each comparison result does and does not establish.

Table 7: Where each parameter family is located in the article.

4 Instructional Rule for Comparison Coordinates

A comparison-layer calculation should follow four steps:

$$\begin{array}{c}\text{fix the CPTG-native branch}\\ \downarrow \\ \text{choose the observational coordinate system}\\ \downarrow \\ \text{apply the locked transport map}\\ \downarrow \\ \text{compare without sector-by-sector retuning.}\end{array} \tag{9}$$

This rule prevents the comparison layer from becoming a hidden fitting layer. If a parameter is allowed to move during a likelihood calculation, it must be identified as a nuisance or control parameter of the observational pipeline, not as a new CPTG branch parameter.

5 Distance and Acoustic-Ruler Comparison Layer

The distance sector begins by expressing the fixed CPTG expansion function in the distance variables used by supernova and BAO pipelines. For a spatially flat comparison branch, the transverse comoving distance is

$$D_M(z) = \frac{c}{H_0}\int_0^z \frac{dz'}{E(z')}. \tag{10}$$

The radial Hubble distance is

$$D_H(z) = \frac{c}{H_0 E(z)}. \tag{11}$$

The volume-averaged acoustic distance is

$$D_V(z) = \left[z D_M(z)^2 D_H(z)\right]^{1/3}. \tag{12}$$

BAO likelihoods do not compare the dimensional distances alone. They compare acoustic ratios. The primary BAO comparison coordinates are

$$\frac{D_M(z)}{r_d}, \tag{13}$$

$$\frac{D_H(z)}{r_d}, \tag{14}$$

and

$$\frac{D_V(z)}{r_d}. \tag{15}$$

For anisotropic BAO or ShapeFit-style comparisons, the same distance information may be expressed through Alcock–Paczynski coordinates. In the parallel and transverse convention,

$$q_{\parallel} = \frac{D_H(z)/r_d}{[D_H(z)/r_d]_{\rm fid}}, \tag{16}$$

$$q_{\perp} = \frac{D_M(z)/r_d}{[D_M(z)/r_d]_{\rm fid}}. \tag{17}$$

The isotropic and anisotropic combinations are then

$$q_{\rm iso} = \left(q_{\perp}^2 q_{\parallel}\right)^{1/3}, \tag{18}$$

and

$$q_{\rm ap} = \frac{q_{\parallel}}{q_{\perp}}. \tag{19}$$

The acoustic ruler is a comparison-layer normalization. When an absolute BAO-ruler normalization is required, the branch value used here is

$$r_d = 148.998491 \text{ Mpc}. \tag{20}$$

The native CPTG object is not a fitted distance point, but the full distance relation

$$\{D_M(z), D_H(z), D_V(z)\}, \tag{21}$$

together with the acoustic-ratio map

$$\left\{\frac{D_M(z)}{r_d}, \frac{D_H(z)}{r_d}, \frac{D_V(z)}{r_d}, q_{\parallel}, q_{\perp}, q_{\rm iso}, q_{\rm ap}\right\}. \tag{22}$$

Supernova comparisons use the luminosity distance

$$D_L(z) = (1+z)D_M(z). \tag{23}$$

The corresponding distance modulus is

$$\mu(z) = 5\log_{10}\left[\frac{D_L(z)}{\text{Mpc}}\right] + 25. \tag{24}$$

The supernova comparison is therefore a distance-shape comparison after standardization, intercept treatment, calibration handling, and covariance weighting [2]. BAO comparisons are made through dimensionless acoustic ratios and AP coordinates defined by the survey likelihood convention [3]. In both cases, CPTG supplies the geometric distance relation first; the observational pipeline determines which comparison coordinates are used.

6 Baryon and Light-Element Conversion

The CPTG baryon fraction is fixed by the geometric relation

$$f_b = \frac{\pi}{20}. \tag{25}$$

Numerically,

$$f_b = 0.157079632679. \tag{26}$$

The acoustic/CMB physical baryon-density comparison value is

$$\Omega_b h^2_{\mathrm{ac}} = 0.0225278574944. \tag{27}$$

The corresponding acoustic/CMB baryon-to-photon comparison value is

$$\eta_{10}^{\mathrm{ac}} = 6.170380167710. \tag{28}$$

The effective-radiation comparison value is identified with the acoustic exponent:

$$N_{\mathrm{eff}}^{\mathrm{CPTG}} = p_{\mathrm{ac}}. \tag{29}$$

Therefore,

$$N_{\mathrm{eff}}^{\mathrm{CPTG}} = 2.968584073464. \tag{30}$$

The light-element comparison layer uses the transport-scaled baryon conversion

$$\eta_{10}^{\mathrm{BBN}} = \eta_{10}^{\mathrm{ac}} \sqrt{\mathcal{G}_T}. \tag{31}$$

This gives

$$\eta_{10}^{\mathrm{BBN}} = 5.998071834744. \tag{32}$$

The corresponding physical baryon-density conversion is

$$\Omega_b h^2_{\mathrm{BBN}} = 0.021898765370. \tag{33}$$

Thus the baryon conversion separates the acoustic/CMB layer from the nuclear-reaction layer:

$$\left(\Omega_b h^2_{\mathrm{ac}}, \eta_{10}^{\mathrm{ac}}\right) \longrightarrow \left(\Omega_b h^2_{\mathrm{BBN}}, \eta_{10}^{\mathrm{BBN}}\right). \tag{34}$$

The light-element reaction-network comparison values are

$$N_{\text{eff}}^{\text{network}} = 2.968611196765, \tag{35}$$
$$Y_p^{\text{network}} = 0.245694338195, \tag{36}$$
$$10^6(\text{D/H})_{\text{network}} = 25.074383114428. \tag{37}$$

The helium and deuterium comparison is therefore controlled by the transported baryon layer, not by direct insertion of the acoustic/CMB baryon density into the nuclear epoch. CPTG supplies the baryon fraction, radiation comparison value, and transport-scaled baryon-to-photon conversion; the light-element network then maps those quantities into

$$Y_p \qquad \text{and} \qquad \text{D/H}. \tag{38}$$

This makes the BBN comparison a transport-layer test of the baryon and radiation sector rather than an independent fit of primordial abundance parameters [7, 8].

7 Perturbation, Growth, and Lensing Channels

CPTG treats growth and lensing as reduced perturbation channels of the nonlinear curvature-polarization and curvature-transport field. In scalar perturbation form, the metric is written as

$$ds^2 = -\left(1 + 2\Psi\right) c^2 dt^2 + a(t)^2 \left(1 - 2\Phi\right) d\mathbf{x}^2. \tag{39}$$

The growth response is defined by the effective Poisson relation

$$k^2\Psi = -4\pi G a^2 \bar{\rho}\, \mu_{\text{CPTG}}(a, k)\delta. \tag{40}$$

The lensing response is defined through the Weyl-potential combination

$$k^2\left(\Phi + \Psi\right) = -8\pi G a^2 \bar{\rho}\, \Sigma_{\text{CPTG}}(a, k)\delta. \tag{41}$$

For the reduced comparison layer used here, the active scale-averaged growth and lensing responses are

$$\mu_{\text{CPTG}}(a) = 1 + \frac{\pi}{2}\left(A_L^{\text{CPTG}} - 1\right)\left(1 - a^{p_{\text{ac}}}\right), \tag{42}$$
$$\Sigma_{\text{CPTG}}(a) = 1 + \left(A_L^{\text{CPTG}} - 1\right)\left(1 - a^{p_{\text{ac}}}\right). \tag{43}$$

The native lensing response scale is fixed by the curvature-to-acoustic exponent ratio:

$$A_L^{\text{CPTG}} = \left(\frac{p_C}{p_{\text{ac}}}\right)^2. \tag{44}$$

With

$$p_C = \pi \tag{45}$$

and

$$p_{\mathrm{ac}} = 3 - \frac{\pi}{100}, \tag{46}$$

this gives

$$A_L^{\mathrm{CPTG}} = 1.119956202138. \tag{47}$$

The raw CMB-lensing bandpower comparison is not the same object as the native lensing response. The projected bandpower amplitude is

$$A_{\mathrm{bandpower}} = 1 + q_{\mathrm{bandpower}} \left(A_L^{\mathrm{CPTG}} - 1 \right), \tag{48}$$

where

$$q_{\mathrm{bandpower}} = \frac{1 - \mathcal{G}_T}{\mathcal{B}_\pi}. \tag{49}$$

Therefore,

$$A_{\mathrm{bandpower}} = 1.009690662776. \tag{50}$$

The late-time weak-lensing comparison amplitude is

$$S_{8,\mathrm{late}}^{\mathrm{CPTG}} = 0.7880713335232. \tag{51}$$

The compressed CMB-lensing amplitude channel is

$$S_{8,\mathrm{CMBL}}^{\mathrm{CPTG}} = S_{8,\mathrm{late}}^{\mathrm{CPTG}} \left(A_L^{\mathrm{CPTG}} \right)^{\alpha_C}. \tag{52}$$

The compression exponent is

$$\alpha_C = 1 - \frac{2}{\pi}. \tag{53}$$

This gives

$$S_{8,\mathrm{CMBL}}^{\mathrm{CPTG}} = 0.821191065949. \tag{54}$$

Thus CPTG distinguishes three lensing-related quantities:

$$A_L^{\mathrm{CPTG}}, \qquad A_{\mathrm{bandpower}}, \qquad S_{8,\mathrm{CMBL}}^{\mathrm{CPTG}}. \tag{55}$$

The first is the native response scale, the second is a projected raw bandpower comparison, and the third is a compressed amplitude comparison. These are different observables and should not be collapsed into a single fitted lensing parameter.

8 Locked CMB Acoustic Transport Layer

The CPTG CMB comparison branch is fixed before likelihood evaluation. The curvature exponent is

$$p_C = \pi, \tag{56}$$

and the acoustic exponent is

$$p_{\rm ac} = 3 - \frac{\pi}{100}. \tag{57}$$

The acoustic transport factor is

$$\mathcal{G}_T = \frac{p_{\rm ac}}{p_C} = \frac{3 - \pi/100}{\pi}. \tag{58}$$

Numerically,

$$\mathcal{G}_T = 0.9449296585513721, \tag{59}$$

with

$$\sqrt{\mathcal{G}_T} = 0.9720749243506758. \tag{60}$$

The cold-sector comparison transport is

$$\mathcal{D}_c = \sqrt{\mathcal{G}_T}\left[1 + \frac{1}{10\pi}\left(\frac{\pi}{3} - 1\right)\right]. \tag{61}$$

Equivalently,

$$\mathcal{D}_c = \sqrt{\mathcal{G}_T}\left(1 + \frac{\pi - 3}{30\pi}\right). \tag{62}$$

Therefore,

$$\mathcal{D}_c = 0.9735353159758136. \tag{63}$$

The scalar-amplitude comparison gate is

$$\mathcal{A}_s = \frac{\pi}{3}, \tag{64}$$

so that

$$\mathcal{A}_s = 1.0471975511965976. \tag{65}$$

The optical-depth comparison gate is the two-step visibility contraction of the acoustic transport branch,

$$\mathcal{T}_\tau = \mathcal{G}_T^2. \tag{66}$$

Thus,

$$\mathcal{T}_\tau = 0.8928920596100127. \tag{67}$$

The locked comparison map is

$$\omega_{c,\text{eff}}h^2 = \omega_{c,\text{CPTG}}h^2\mathcal{D}_c, \tag{68}$$
$$A_{s,\text{eff}} = A_{s,\text{CPTG}}\mathcal{A}_s, \tag{69}$$
$$\tau_{\text{eff}} = \tau_{\text{CPTG}}\mathcal{G}_T^2. \tag{70}$$

Using the native comparison inputs

$$\omega_{c,\text{CPTG}}h^2 = 0.120885025627, \tag{71}$$
$$A_{s,\text{CPTG}} = 2.04 \times 10^{-9}, \tag{72}$$
$$\tau_{\text{CPTG}} = 0.066, \tag{73}$$

the effective CMB comparison values are

$$\omega_{c,\text{eff}}h^2 = 0.11768584162052577, \tag{74}$$
$$A_{s,\text{eff}} = 2.1362830044410592 \times 10^{-9}, \tag{75}$$
$$\tau_{\text{eff}} = 0.05893087593426084. \tag{76}$$

No phenomenological lensing enhancement is applied:

$$A_{\text{lens}} = 1. \tag{77}$$

The complete locked CMB comparison row is

Parameter	Locked value
H_0	67.4777967351 $\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$
$\omega_b h^2$	0.022527857494
$\omega_c h^2$	0.117685841620526
n_s	0.968584073464
N_{eff}	2.968584073464
Y_{He}	0.2485
A_s	$2.136283004441 \times 10^{-9}$
τ	0.058930875934
A_{lens}	1

Table 8: Locked CPTG CMB acoustic-transport comparison row.

This row is the fixed CMB acoustic-transport comparison layer. It is not a parameter fit, and it is not adjusted separately for individual likelihood channels.

9 Geometric Origin of the CMB Amplitude and Visibility Gates

The locked CMB comparison layer contains two geometric projection gates beyond the cold-sector distance map: the scalar-amplitude gate and the optical-depth visibility gate. These are not fitted parameters, and they are not borrowed from an external cosmological model. They are CPTG curvature projections that translate native curvature transport into the acoustic amplitude and visibility coordinates measured by the CMB sky.

The governing curvature branch is

$$p_C = \pi. \tag{78}$$

The acoustic transport branch is

$$p_{\mathrm{ac}} = 3 - \frac{\pi}{100}. \tag{79}$$

The corresponding acoustic-to-curvature transport ratio is

$$\mathcal{G}_T = \frac{p_{\mathrm{ac}}}{p_C}. \tag{80}$$

The purpose of the comparison layer is therefore narrow: CPTG-native curvature quantities are projected into acoustic observables, visibility suppression, and lensing reconstruction coordinates. The theory remains CPTG geometry throughout the mapping.

9.1 Amplitude Projection and the $\pi/3$ Curvature Gate

The scalar amplitude in the CMB comparison layer is the two-point scalar projection of a CPTG curvature-response field. The native curvature branch is organized by $p_C = \pi$, while the acoustic scalar normalization is a three-spatial-direction projection. The branch-to-acoustic projection is therefore

$$\mathcal{A}_C = \frac{p_C}{3}. \tag{81}$$

Since

$$p_C = \pi, \tag{82}$$

the amplitude projection becomes

$$\mathcal{A}_C = \frac{\pi}{3}. \tag{83}$$

The effective scalar amplitude used in the CMB comparison layer is therefore

$$A_{s,\mathrm{eff}} = A_{s,\mathrm{CPTG}}\mathcal{A}_C. \tag{84}$$

Substituting the geometric amplitude projection gives

$$A_{s,\mathrm{eff}} = A_{s,\mathrm{CPTG}}\frac{\pi}{3}. \tag{85}$$

For the locked CPTG-native amplitude,

$$A_{s,\mathrm{CPTG}} = 2.04 \times 10^{-9}, \tag{86}$$

the projected comparison-layer amplitude is

$$A_{s,\mathrm{eff}} = 2.04 \times 10^{-9} \frac{\pi}{3}. \tag{87}$$

Thus,

$$A_{s,\mathrm{eff}} = 2.136283004441 \times 10^{-9}. \tag{88}$$

This value is not introduced as an independent normalization. It is the scalar CMB projection of the CPTG curvature-amplitude branch.

9.2 Visibility Transport and the $\mathcal{G}_T^2$ Optical-Depth Gate

The optical-depth comparison coordinate measures visibility-layer suppression of the observed CMB amplitude. In the locked CPTG comparison row, this channel is not retuned against low-ℓ polarization. It is carried by the second-order acoustic transport visibility projection

$$\mathcal{T}_\tau = \mathcal{G}_T^2. \tag{89}$$

The effective optical depth in the CMB comparison layer is therefore

$$\tau_{\mathrm{eff}} = \tau_{\mathrm{CPTG}} \mathcal{T}_\tau. \tag{90}$$

For the locked native value

$$\tau_{\mathrm{CPTG}} = 0.066, \tag{91}$$

one obtains

$$\tau_{\mathrm{eff}} = 0.066 \times \mathcal{G}_T^2 = 0.05893087593426084. \tag{92}$$

The optical-depth gate is therefore a fixed acoustic-transport visibility projection inside the CMB comparison coordinates. It is not an empirical retuning of optical depth.

9.3 Locked Gate Closure

The CMB comparison row closes through three CPTG geometric maps:

$$\omega_{c,\mathrm{eff}} h^2 = \omega_{c,\mathrm{CPTG}} h^2 \mathcal{D}_c, \tag{93}$$

$$A_{s,\mathrm{eff}} = A_{s,\mathrm{CPTG}} \frac{\pi}{3}, \tag{94}$$

$$\tau_{\mathrm{eff}} = \tau_{\mathrm{CPTG}} \mathcal{G}_T^2. \tag{95}$$

The curvature-distance gate $\mathcal{D}_c$ is already fixed by the CPTG comparison geometry. The amplitude and visibility gates now have direct geometric interpretations: $\pi/3$ is the scalar acoustic projection of the π-curvature branch, while $\mathcal{T}_\tau = \mathcal{G}_T^2$ is the two-step visibility contraction of the acoustic transport branch.

The locked effective values are therefore

$$\omega_{c,\mathrm{eff}} h^2 = 0.117685841620526, \tag{96}$$

$$A_{s,\mathrm{eff}} = 2.136283004441 \times 10^{-9}, \tag{97}$$

$$\tau_{\mathrm{eff}} = 0.058930875934. \tag{98}$$

The lensing amplitude remains fixed at

$$A_{\mathrm{lens}} = 1. \tag{99}$$

No lensing inflation is introduced. No posterior retuning is introduced. The locked CMB comparison layer is therefore a geometric projection of CPTG curvature transport into acoustic amplitude, visibility, matter-density, and lensing coordinates.

10 Cleaned CMB Verification and Full-Stack Validation

The cleaned CMB branch is evaluated with fixed CPTG cosmology and fixed comparison gates. The comparison statistic is

$$\Delta\chi^2 = \chi^2_{\mathrm{CPTG,locked}} - \chi^2_{\mathrm{control}}, \tag{100}$$

where the control row is evaluated under the same likelihood, profiling protocol, posterior protocol, and data combination. Control-level recovery is defined by

$$\Delta\chi^2 \le 5. \tag{101}$$

A decisive stress level is defined by

$$\Delta\chi^2 > 10. \tag{102}$$

All CMB comparisons in this section keep the locked gates fixed:

$$\mathcal{D}_c = 0.9735353159758136, \tag{103}$$

$$\mathcal{A}_s = \frac{\pi}{3}, \tag{104}$$

$$\mathcal{T}_\tau = \mathcal{G}_T^2, \tag{105}$$

and

$$A_{\mathrm{lens}} = 1. \tag{106}$$

No result in this section retunes the cold-sector transport, scalar-amplitude gate, optical-depth gate, or lensing-amplitude setting.

10.1 Validation Milestone Summary

The CMB comparison layer is tested through high-ℓ likelihoods, posterior-proxy sampling, low-ℓ polarization, lensing diagnostic channels, and combined likelihood stacks. Table 9 summarizes the current validation status of the locked cleaned $\mathcal{G}_T^2$ comparison row.

The low-ℓ and lensing row is labeled as a diagnostic pass because it tests the fixed cleaned comparison row in those channels alone; the validation claim is made only after this diagnostic layer is combined with the high-ℓ, profile-likelihood, posterior, and gate-closure validation layers.

Validation layer	Decisive result	Status
High-ℓ profile robustness	Plik TTTEEE, SPT-3G TEEE, NPIPE CamSpec TTTEEE: mean median shift +2.221794; maximum median shift +4.501564; CamSpec worst repeat +5.013677.	Validation pass; CamSpec watch
Converged posterior-proxy check	Plik TTTEEE and SPT-3G TEEE: mean shift +1.6722; maximum shift +1.9233.	Validation pass
Low-ℓ and lensing diagnostic stability	Planck low-ℓ EE, Planck lensing, ACT DR6 lensing: Planck mean +1.589; maximum Planck shift +2.149.	Diagnostic pass
Full-stack profile validation	Plik+lowE+lensing, NPIPE CamSpec+lowE+lensing, SPT+ACT: mean primary shift +2.101464; maximum primary shift +3.669953; CamSpec worst repeat +5.990520.	Validation pass
Combined-stack posterior validation	Plik+lowE+lensing, full NPIPE CamSpec+lowE+lensing, SPT+ACT: mean median shift +0.096642; maximum median shift +0.737534; maximum recovered $R-1 = 0.143463$.	Validation pass
Final validation consolidation	Consolidated profile, posterior, convergence, and gate status: maximum $\lvert\Delta\chi^2\rvert = 4.824743$ with $A_{\rm lens} = 1$.	Validation pass
Amplitude/tau gate closure	Gates $\mathcal{A}_s = \pi/3$ and $\mathcal{T}_\tau = \mathcal{G}_T^2$: exact closure with maximum relative difference 0.	Numerical validation pass

Table 9: Current CMB validation milestones for the locked cleaned $\mathcal{G}_T^2$ comparison row.

10.2 High-ℓ Profile Robustness

The locked cleaned row is tested against independent high-ℓ likelihood implementations using repeated nuisance-profile comparisons. The aggregate decisive median shift is

$$\left\langle \Delta\chi^2_{\rm median}\right\rangle = 2.221794, \tag{107}$$

and the maximum decisive high-ℓ median shift is

$$\max\left(\Delta\chi^2_{\rm median}\right) = 4.501564. \tag{108}$$

The Plik and SPT-3G channels are stable at control level. NPIPE CamSpec remains control-level by the comparison statistic while showing profile sensitivity associated with nuisance and calibration basin structure in the full CamSpec implementation.

10.3 Plik and SPT Posterior-Proxy Robustness

The decisive Plik and SPT-3G high-ℓ channels are also tested with MCMC posterior-proxy sampling. The chains converge before the sample cap in all four decisive chains. The result is summarized in Table 10.

Target	CPTG proxy χ^2	Control proxy χ^2	Shift	CPTG samples	Status
Plik TTTEEE	2344.4088	2342.4855	+1.9233	33520	Control-level
SPT-3G TEEE	1112.9661	1111.5450	+1.4211	18680	Control-level

Table 10: Converged Plik/SPT posterior-proxy comparison for the locked cleaned row.

The corresponding control chains contain 57600 accepted Plik samples and 28120 accepted SPT samples. The final mean $R-1$ values are

$$R_{\text{Plik,CPTG}} - 1 = 0.029169, \qquad R_{\text{Plik,control}} - 1 = 0.029464, \tag{109}$$
$$R_{\text{SPT,CPTG}} - 1 = 0.028239, \qquad R_{\text{SPT,control}} - 1 = 0.025843. \tag{110}$$

The mean decisive posterior-proxy shift is

$$\left\langle \Delta\chi^2_{\text{posterior}} \right\rangle = 1.6722, \tag{111}$$

with maximum decisive posterior-proxy shift

$$\max\left(\Delta\chi^2_{\text{posterior}}\right) = 1.9233. \tag{112}$$

The Plik and SPT posterior-proxy channels therefore remain inside the control-level region under the locked CMB comparison row.

10.4 Low-ℓ and Lensing Diagnostic Stability

The cleaned row is tested in low-ℓ polarization and lensing channels without retuning τ and without increasing A_{lens}. Table 11 reports the diagnostic stability result.

Target	CPTG median χ^2	Control median χ^2	Median shift	Status
Planck low-ℓ EE	396.550	395.521	+1.029	Control-level
Planck 2018 lensing	10.936	8.88949	+2.149	Control-level
ACT DR6 lensing	14.7052	13.9677	+0.7375	Control-level

Table 11: Low-ℓ and lensing diagnostic stability for the locked cleaned row.

For the decisive Planck low-ℓ and Planck lensing rows, the mean median shift is

$$\left\langle \Delta\chi^2_{\mathrm{lowE/lensing}} \right\rangle = 1.589, \tag{113}$$

and the maximum decisive median shift is

$$\max\left(\Delta\chi^2_{\mathrm{lowE/lensing}}\right) = 2.149. \tag{114}$$

Thus the cleaned optical-depth relation

$$\tau_{\mathrm{eff}} = 0.05893087593426084 \tag{115}$$

remains compatible with the low-ℓ and lensing diagnostic channels under the same locked comparison layer.

10.5 Combined-Stack Profile Validation

The strongest completed CMB validation layer is the locked combined-stack profile audit. This test combines high-ℓ, low-ℓ, and lensing likelihoods where available and keeps the same locked CMB row. The audit treats NPIPE CamSpec as a decisive Planck PR4 pipeline. When nuisance-marginalized CamSpec-NPIPE-lite is not available in the likelihood environment, full NPIPE CamSpec is evaluated with a best-of-N profile-sensitive protocol [16].

All enabled combined-stack profile runs succeed:

$$44/44 = 1. \tag{116}$$

There are no timeouts and no decisive run failures. The decisive combined-stack results are given in Table 12.

Stack	Protocol	Primary shift	Median shift	Worst shift
Plik TTTEEE + lowE + Planck lensing	Median profile	+1.896906	+1.896906	+2.418011
NPIPE CamSpec + lowE + Planck lensing	Best-of-N profile	+3.669953	+4.824743	+5.990520
SPT-3G TEEE + ACT DR6 lensing	Median profile	+0.737534	+0.737534	+0.737534

Table 12: Locked combined-stack CMB profile validation. NPIPE CamSpec is treated as a decisive PR4 pipeline using full CamSpec best-of-N when CamSpec-NPIPE-lite is unavailable.

The decisive primary-shift mean is

$$\left\langle \Delta\chi^2_{\mathrm{primary}} \right\rangle = 2.101464, \tag{117}$$

and the largest decisive primary shift is

$$\max\left(\Delta\chi^2_{\mathrm{primary}}\right) = 3.669953. \tag{118}$$

This places Plik, decisive NPIPE CamSpec, and the ground high-ℓ+lensing stack inside the control-level region under the same fixed CMB row. The CamSpec worst repeat exceeds 5 but remains below the decisive stress threshold, and the primary best-of-N CamSpec comparison remains control-level. This is interpreted as successful combined-stack profile validation with CamSpec retained as a decisive but profile-sensitive PR4 likelihood pipeline.

10.6 Combined-Stack Posterior Validation

The locked combined-stack posterior validation applies the same cleaned CMB comparison row to posterior sampling across Plik, decisive NPIPE CamSpec, and SPT/ACT stack combinations. The validation keeps

$$A_{\mathrm{lens}} = 1. \tag{119}$$

The posterior layer does not retune τ, $\mathcal{D}_c$, $\mathcal{A}_s$, or $\mathcal{T}_\tau$. All enabled MCMC runs succeed:

$$6/6 = 1. \tag{120}$$

There are no timeouts and no decisive run failures. The CamSpec-lite probe does not select an installed nuisance-marginalized component, so full NPIPE CamSpec is retained as the decisive CamSpec comparison under the profile-sensitive posterior protocol.

Stack	Median shift	Max final $R-1$	Status
Plik TTTEEE + lowE + Planck lensing	−0.2238004	0.121641	Control-level
NPIPE CamSpec + lowE + Planck lensing	−0.2238066	0.143463	Control-level
SPT-3G TEEE + ACT DR6 lensing	+0.7375340	0.108998	Control-level

Table 13: Locked combined-stack CMB posterior validation. Full NPIPE CamSpec is used decisively when CamSpec-lite is unavailable.

The decisive median-shift mean is

$$\left\langle \Delta\chi^2_{\mathrm{combined\ posterior}} \right\rangle = 0.0966423333, \tag{121}$$

and the largest decisive median shift is

$$\max\left(\Delta\chi^2_{\mathrm{combined\ posterior}}\right) = 0.737534. \tag{122}$$

The maximum final convergence value across decisive posterior chains is

$$\max(R-1) = 0.143463. \tag{123}$$

The combined-stack posterior decision is therefore

$$\text{LOCKED\_CPTG\_COMBINED\_POSTERIOR\_CONTROL\_LEVEL.} \tag{124}$$

This closes the combined-stack posterior validation layer for the locked cleaned CMB comparison row while preserving $A_{\rm lens} = 1$ and the fixed acoustic-transport gates.

10.7 Numerical Amplitude and Optical-Depth Gate Closure

The locked comparison gates are also checked as an explicit closure calculation against the final CMB row. This audit does not rerun any likelihood, does not change the CMB row, and does not alter the validation protocol. It verifies the arithmetic closure of the native-to-effective map,

$$\omega_{c,\rm eff}h^2 = \omega_{c,\rm CPTG}h^2\mathcal{D}_c, \tag{125}$$

$$A_{s,\rm eff} = A_{s,\rm CPTG}\frac{\pi}{3}, \tag{126}$$

$$\tau_{\rm eff} = \tau_{\rm CPTG}\mathcal{G}_T^2. \tag{127}$$

Using the locked native values,

$$\omega_{c,\rm CPTG}h^2 = 0.120885025627, \tag{128}$$

$$A_{s,\rm CPTG} = 2.04 \times 10^{-9}, \tag{129}$$

$$\tau_{\rm CPTG} = 0.066, \tag{130}$$

the closure audit gives

$$0.120885025627\,\mathcal{D}_c = 0.1176858416205258, \tag{131}$$

$$2.04 \times 10^{-9}\left(\frac{\pi}{3}\right) = 2.136283004441059 \times 10^{-9}, \tag{132}$$

$$0.066\left(\mathcal{G}_T^2\right) = 0.05893087593426084. \tag{133}$$

These are exactly the locked effective values used in the CMB comparison row. The maximum relative closure difference is

$$0. \tag{134}$$

The gate-closure decision is

$$\begin{gathered}\text{LOCKED\_CMB\_GATE\_CLOSURE}\\ \text{NUMERICAL\_PASS}\\ \text{GEOMETRIC\_GATE\_STATED.}\end{gathered} \tag{135}$$

The result closes the numerical bookkeeping of the scalar-amplitude and optical-depth gates. The preceding geometric section states these gates as CPTG curvature-response projections: $\mathcal{A}_s = \pi/3$ is the scalar projection of the π-curvature branch, and $\mathcal{T}_\tau = \mathcal{G}_T^2$ is the two-step visibility contraction of the acoustic transport branch.

10.8 Coupled Gate Perturbation

A coupled $p_{\rm ac}$-gate perturbation tests whether $\mathcal{G}_T^2$ behaves as a coherent transport relation rather than as an isolated numerical value. The perturbation varies $p_{\rm ac}$ around the locked value, holds n_s and $N_{\rm eff}$ fixed to isolate the gate-coupling layer, and applies

$$\mathcal{G}_T(p_{\rm ac}) = \frac{p_{\rm ac}}{\pi}, \tag{136}$$

$$\mathcal{D}_c(p_{\rm ac}) = \sqrt{\mathcal{G}_T(p_{\rm ac})}\left[1 + \frac{1}{10\pi}\left(\frac{\pi}{3} - 1\right)\right], \tag{137}$$

$$\tau_{\rm eff}(p_{\rm ac}) = \tau_{\rm CPTG}\mathcal{G}_T(p_{\rm ac})^2. \tag{138}$$

The result is summarized in Table 14.

Rank	$p_{\rm ac}$ shift	Mean target-median shift	Max target-median shift	Status
1	+0.005	+2.212364	+4.574011	Control-level
2	0.000, locked	+2.297587	+4.677919	In basin
3	+0.010	+2.299781	+4.901645	In basin
4	−0.005	+2.584480	+5.078886	Mild stress
5	−0.010	+3.193148	+5.921761	Stress
6	+0.020	+3.543665	+7.271139	Stress
7	−0.020	+5.479216	+9.497230	Poor

Table 14: Coupled $p_{\rm ac}$-gate perturbation. The locked row remains inside the coupled-gate basin.

The locked row ranks second and is only

$$0.085223 \tag{139}$$

above the best aggregate row in the perturbation audit. This supports $\mathcal{G}_T^2$ as a coherent coupled-gate relation inside the locked CMB acoustic-transport layer.

10.9 Current CMB Status

The current CMB status is

$$\begin{gathered}\text{CPTG passes the locked cleaned CMB comparison layer}\\ \text{through high-}\ell\text{ profile robustness,}\\ \text{converged Plik/SPT posterior-proxy tests,}\\ \text{low-}\ell\text{/lensing diagnostic stability,}\\ \text{and combined-stack profile and posterior validation}\\ \text{across Plik, decisive NPIPE CamSpec,}\\ \text{and SPT/ACT, with } A_{\rm lens} = 1.\end{gathered} \tag{140}$$

The combined-stack posterior layer is included as a completed control-level validation claim for the locked cleaned CMB comparison row. The completed CMB evidence supports the locked row without optical-depth retuning or phenomenological lensing enhancement. The decisive posterior chains have maximum $R - 1 = 0.143463$, and the gate-closure audit verifies exact numerical closure of $\omega_{c,\text{eff}} h^2$, $A_{s,\text{eff}}$, and τ_{eff} against the locked row.

11 CMB Comparison Coordinates and Locked CPTG Quantities

The locked CPTG CMB row is a CPTG-native geometric construction expressed in the coordinates used by CMB likelihoods. The row is obtained by mapping native branch quantities through fixed acoustic-transport gates into measurable CMB comparison coordinates. These coordinates are used only for comparison. They do not define the theory.

Quantity	CPTG locked value	Control value	Difference	Ratio
H_0	67.4777967351	67.36	+0.1177967351	1.001749
$\omega_b h^2$	0.022527857494	0.02237	+0.000157857494	1.007057
$\omega_c h^2$	0.117685841621	0.1200	−0.002314158379	0.980715
n_s	0.968584073464	0.9649	+0.003684073464	1.003818
N_{eff}	2.968584073464	3.046	−0.077415926536	0.974584
Y_{He}	0.2485	0.2454	+0.0031	1.012632
A_s	$2.136283004441 \times 10^{-9}$	2.100549×10^{-9}	$+3.5734004441 \times 10^{-11}$	1.017012
τ	0.058930875934	0.0544	+0.004530875934	1.083288

Table 15: Locked CPTG CMB row expressed in likelihood comparison coordinates with a local control row.

The largest comparison-layer shifts occur in the optical depth, effective radiation value, cold physical density, and scalar amplitude. These are not independent fitted offsets. They are the quantities acted on by the locked CPTG acoustic-transport map,

$$\omega_{c,\text{eff}} h^2 = \omega_{c,\text{CPTG}} h^2 \mathcal{D}_c, \tag{141}$$

$$A_{s,\text{eff}} = A_{s,\text{CPTG}} \frac{\pi}{3}, \tag{142}$$

$$\tau_{\text{eff}} = \tau_{\text{CPTG}} \mathcal{G}_T^2. \tag{143}$$

The comparison should therefore be read as

$$\begin{aligned} \text{CMB likelihood coordinates} = {} & \text{CPTG-native quantities} \\ & \text{mapped through fixed acoustic-transport gates.} \end{aligned} \tag{144}$$

It should not be read as

$$\text{CPTG-native quantities} = \text{externally inferred comparison coordinates.} \tag{145}$$

The role of the control row is to provide a local comparison baseline under the same likelihood machinery. The CPTG row is judged by the resulting likelihood shifts, not by requiring each mapped coordinate to equal the control coordinate exactly.

12 Compressed Validation Suite Results

The fixed branch can be compared against compressed observational sectors using explicit coordinate conversions. The comparisons below use the native quantities stated in this article. They are not independent parameter refits.

12.1 Pantheon+ Supernova Distance-Shape Test

The Pantheon+ comparison uses the full statistical-plus-systematic covariance matrix and analytically marginalizes one intercept. The primary sample removes calibrator rows, so the comparison tests the shape of the supernova distance-redshift relation rather than an absolute H_0 calibration [2].

Model	χ^2	Reduced χ^2	$\Delta\chi^2$
Planck-like ΛCDM	1458.014422	0.898345	0
CPTG raw π branch	1458.058573	0.898373	+0.044151
CPTG acoustic/clock branch	1458.644632	0.898734	+0.630210
DESI-like ΛCDM	1460.490455	0.899871	+2.476033

Table 16: Pantheon+ full-covariance supernova distance-shape comparison. The primary mode uses $N = 1624$ non-calibrator supernova rows and 1623 effective degrees of freedom after intercept marginalization.

The supernova sector therefore gives a distance-shape pass for the locked CPTG branch. The result is not an H_0 calibration claim, because the intercept is marginalized and calibrator rows are excluded from the primary shape comparison.

12.2 Age and Clock Consistency

The raw geometric age and transported clock age are distinct CPTG layers. The raw branch reports the native geometric clock, while the transported branch gives the clock comparison map associated with the acoustic exponent.

Branch	t_0 Gyr	Difference	Ratio
CPTG raw π geometric	12.754121	-1.026208	0.925531
CPTG transported clock $p_{\rm ac}$	13.885173	$+0.104845$	1.007608
CPTG acoustic-density clock	13.926455	$+0.146127$	1.010604
Planck-like ΛCDM	13.780328	0	1
DESI-like ΛCDM	13.970356	$+0.190028$	1.013790

Table 17: Present-age comparison across CPTG and standard comparison branches.

The transported CPTG clock branch gives a coherent present-age comparison and lies within about one percent of the Planck-like comparison age. This is a clock-map result, not a replacement of the native geometric age layer.

12.3 BBN Primary-Abundance Comparison

The BBN comparison uses the transported baryon density rather than direct insertion of the acoustic/CMB baryon density into the nuclear reaction epoch. The comparison-layer inputs are

$$\eta_{10}^{\rm BBN} = 5.998071834744, \tag{146}$$

$$\Omega_b h^2_{\rm BBN} = 0.021898765370, \tag{147}$$

$$N_{\rm eff}^{\rm CPTG} = 2.968584073464. \tag{148}$$

Quantity	CPTG value	Comparison value	Difference	Significance
Y_p	0.245694338195	0.245 ± 0.003	$+0.000694338195$	$+0.231446\sigma$
10^6(D/H)	25.074383114428	25.47 ± 0.29	-0.395616885572	-1.364196σ
$N_{\rm eff}$	2.968584073464	3.044 ± 0.18	-0.075415926536	-0.418977σ

Table 18: BBN primary-abundance comparison for the locked CPTG branch.

The maximum primary-abundance deviation is

$$1.364196\sigma. \tag{149}$$

The locked branch therefore passes the primary BBN comparison within the representative two-sigma band [7, 8]. Lithium is not used as a pass/fail quantity without a locked CPTG lithium prediction.

12.4 Compressed Weak-Lensing and CMB-Lensing S_8 Comparison

The compressed growth/lensing comparison keeps the late weak-lensing channel and the CMB-lensing channel distinct:

$$S_{8,\mathrm{late}}^{\mathrm{CPTG}} = 0.788071335232, \tag{150}$$
$$S_{8,\mathrm{CMBL}}^{\mathrm{CPTG}} = 0.821191065949. \tag{151}$$

Anchor	CPTG residual	Status
DES Y3 fiducial	$+1.211306\sigma$	Within 2σ
DES Y3 optimized scale cuts	$+0.945373\sigma$	Within 2σ
KiDS-1000	$+1.292059\sigma$	Within 2σ
HSC-Y3	$+0.371426\sigma$	Within 2σ
Weighted weak-lensing mean	$+1.941517\sigma$	Within 2σ

Table 19: Compressed weak-lensing comparison for $S_{8,\mathrm{late}}^{\mathrm{CPTG}} = 0.788071335232$.

For the weighted weak-lensing mean, the comparison value is

$$S_{8,\mathrm{WL}}^{\mathrm{weighted}} = 0.766523351546, \tag{152}$$

with

$$\sigma_{S_8,\mathrm{WL}} = 0.011098531049. \tag{153}$$

The CPTG late-time value differs by

$$S_{8,\mathrm{late}}^{\mathrm{CPTG}} - S_{8,\mathrm{WL}}^{\mathrm{weighted}} = 0.021547983686, \tag{154}$$

which is

$$1.941517\sigma. \tag{155}$$

For the CMB-lensing comparison channel, the weighted CMB anchor is

$$S_{8,\mathrm{CMB}}^{\mathrm{weighted}} = 0.832795294118, \tag{156}$$

with

$$\sigma_{S_8,\mathrm{CMB}} = 0.010089482002. \tag{157}$$

The CPTG CMB-lensing comparison value is

$$S_{8,\mathrm{CMBL}}^{\mathrm{CPTG}} = 0.821191065949. \tag{158}$$

The difference is

$$S_{8,\mathrm{CMBL}}^{\mathrm{CPTG}} - S_{8,\mathrm{CMB}}^{\mathrm{weighted}} = -0.011604228169, \tag{159}$$

or

$$-1.150131\sigma. \tag{160}$$

The compressed S_8 result is a diagnostic pass within representative two-sigma bands [9, 10, 11, 1]. It is not a full shear-likelihood validation.

13 DESI DR1 Compressed-Coordinate, BAO Quarter-Ruler, and Full-Shape Diagnostics

The DESI DR1 coordinate comparison is organized into three distinct layers. The DESI full-shape galaxy-clustering and cosmological-constraint analyses provide the published context for these coordinate-level diagnostics [4, 5]. The first layer is the compressed ShapeFit coordinate comparison using the official DESI DR1 HDF5 likelihood containers associated with the full-shape and BAO clustering products [6]. The second layer is the BAO quarter-ruler coordinate-wrapper diagnostic. The third layer is a full-shape spectrum-shell AP/growth diagnostic using `desilike` coordinate behavior and the native HDF5 spectrum vectors.

The third layer is kept separate from the validation claims. It is an exploratory spectrum-shell diagnostic, not a raw CPTG full-shape power-spectrum likelihood validation. A full raw full-shape validation requires official nuisance-preserving AP, RSD, tracer-window, and covariance wiring.

13.1 Compressed ShapeFit Coordinate Comparison

The ShapeFit HDF5 containers are checked with two reference vectors. The identity vector, where the model vector equals the observed compressed vector, gives

$$\chi^2_{\text{identity}} = 0. \tag{161}$$

The no-shift reference vector is

$$(q_{\text{iso}}, q_{\text{ap}}, d_m, d_f) = (1, 1, 0, 1), \tag{162}$$

which gives

$$\chi^2_{\text{no shift}} = 38.70438441. \tag{163}$$

The locked CPTG compressed-coordinate map converts the transverse and radial AP coordinates into the DESI ShapeFit basis by

$$q_{\text{iso}} = (q_\perp^2 q_\parallel)^{1/3}, \tag{164}$$

and

$$q_{\text{ap}} = \frac{q_\parallel}{q_\perp}. \tag{165}$$

The ShapeFit growth coordinate is used as a ratio to the fiducial growth rate,

$$d_f = \frac{f_{\text{CPTG}}}{f_{\text{fid}}}. \tag{166}$$

The tilt coordinate is held fixed at

$$d_m = 0. \tag{167}$$

Thus the locked compressed ShapeFit vector is

$$\mathbf{x}_{\text{CPTG}} = \left((q_\perp^2 q_\parallel)^{1/3}, \frac{q_\parallel}{q_\perp}, 0, \frac{f_{\text{CPTG}}}{f_{\text{fid}}} \right). \tag{168}$$

For the DESI-like fiducial comparison, the direct HDF5 adapter gives

$$\chi^2_{\mathrm{CPTG}} = 35.19040980558847, \tag{169}$$

with

$$\chi^2_\nu = 1.466267075232853, \tag{170}$$

and

$$\Delta\chi^2 = -3.51397460441153 \tag{171}$$

relative to the no-shift reference. For the Planck-like fiducial cross-check, the corresponding result is

$$\chi^2_{\mathrm{CPTG}} = 38.46800119, \tag{172}$$

$$\chi^2_\nu = 1.60283338, \tag{173}$$

and

$$\Delta\chi^2 = -0.23638322. \tag{174}$$

This is a compressed coordinate-level pass. It is not a raw DESI power-spectrum likelihood validation.

13.2 BAO Quarter-Ruler Coordinate Wrapper

The BAO diagnostic uses the quarter-ruler transport relation

$$\mathcal{G}_T = \frac{p_{\mathrm{ac}}}{p_C}. \tag{175}$$

The corresponding ruler ratio is

$$\frac{r_{d,\mathrm{comp}}}{r_{d,\mathrm{fid}}} = \mathcal{G}_T^{-1/4} = 1.014261942163, \tag{176}$$

with the no-distance-shift AP compression scale

$$Q_\downarrow = \mathcal{G}_T^{1/4} = 0.985938600700. \tag{177}$$

At unity, the BAO compression wrapper gives

$$\chi^2_{\mathrm{BAO}}(q = 1) = 20.12426210852749. \tag{178}$$

The official BAO wrapper exposes Ω_m, $\log \mathcal{L}$, and $\log \pi$ as runtime or pipeline parameters, but it does not expose $q_{\rm iso}$, $q_{\|}$, or $q_{\perp}$ as direct runtime inputs. Therefore the non-unity AP/ruler response is treated as a coordinate-wrapper diagnostic unless an upstream BAO theory calculator supplies the non-unity coordinates through the official likelihood path.

With the CPTG acoustic value of Ω_m and a free ruler ratio, the best non-boundary coordinate-wrapper result occurs near

$$\frac{r_{d,\rm comp}}{r_{d,\rm fid}} \simeq 1.014, \tag{179}$$

with

$$\chi^2 = 13.501774995995627. \tag{180}$$

The locked quarter-ruler value gives

$$\chi^2_{\rm BAO}(\mathcal{G}_T^{-1/4}) = 13.505973446003. \tag{181}$$

An exponent scan using

$$r_d \mapsto \mathcal{G}_T^{\alpha} r_d \tag{182}$$

finds

$$\alpha_{\rm best} = -0.2435, \tag{183}$$

with

$$\chi^2_{\rm best} = 13.501376842627. \tag{184}$$

The CPTG quarter value is

$$\alpha_{\rm CPTG} = -\frac{1}{4}, \tag{185}$$

and is only

$$\Delta\chi^2 = 0.0045966 \tag{186}$$

above the grid best. The quarter-ruler relation is therefore a strong coordinate-wrapper diagnostic and a serious derivation target. It is not presented as uniquely selected by the DESI BAO data alone, and it is not stated as a raw official non-unity q runtime likelihood response.

Layer	Quantity	Result	Status
ShapeFit identity	χ^2	0	Native null pass
ShapeFit no-shift	χ^2	38.70438441	Baseline
ShapeFit CPTG, DESI-like fiducial	χ^2	35.19040980558847	Coordinate-level pass
ShapeFit CPTG, Planck-like fiducial	χ^2	38.46800119	Cross-check
BAO unity wrapper	χ^2	20.12426210852749	Wrapper baseline
BAO best ruler scan	χ^2	13.501774995995627	Non-boundary diagnostic
BAO quarter ruler	χ^2	13.505973446003	Strong coordinate-wrapper diagnostic

Table 20: DESI compressed-coordinate and BAO coordinate-wrapper status. The BAO quarter-ruler row is a coordinate-wrapper diagnostic, not a raw official non-unity runtime response.

13.3 Full-Shape Spectrum AP/Growth Diagnostic

The full-shape spectrum-shell diagnostic evaluates whether the compressed-coordinate success extends into AP/growth spectrum coordinates. This diagnostic uses the native HDF5 spectrum vectors and `desilike` coordinate behavior. It does not inject a CPTG raw matter power spectrum, does not refit DESI nuisance parameters, and does not constitute a DESI full-shape validation.

The `desilike` AP definitions are

$$q_{\parallel} = \frac{(D_H/r_d)}{(D_H/r_d)_{\rm fid}}, \tag{187}$$

and

$$q_{\perp} = \frac{(D_M/r_d)}{(D_M/r_d)_{\rm fid}}. \tag{188}$$

The associated AP and isotropic combinations are

$$q_{\rm ap} = \frac{q_{\parallel}}{q_{\perp}}, \tag{189}$$

and

$$q_{\rm iso} = q_{\parallel}^{\eta} q_{\perp}^{1-\eta}. \tag{190}$$

In isotropic mode,

$$q_{\parallel} = q_{\perp} = q_{\rm iso}. \tag{191}$$

The `BAOPowerSpectrumTemplate` default AP mode is the $q_{\parallel}, q_{\perp}$ convention. The HDF5 BAO products are mixed by tracer: BGS and QSO use $q_{\rm iso}$-type products, while ELG and LRG use $q_{\parallel}, q_{\perp}$-type products. The ShapeFit d_f coordinate scales the growth-rate ratio rather than inserting an absolute $f\sigma_8$.

The spectrum-shell diagnostics show that the locked quarter-down AP plus CPTG growth ratio improves the global spectrum-shell comparison, but the response is tracer- and multipole-dependent. The full sample improves by

$$\Delta(-2\log\mathcal{L}) = -184.04148385539065 \tag{192}$$

for the locked global quarter-down-plus-growth case. When split/tracer convention models are allowed, the best spectrum-shell diagnostic model is the BGS/ELG axis-repair plus LRG/QSO locked model, with

$$\Delta(-2\log\mathcal{L}) = -248.077305. \tag{193}$$

This means the spectrum-shell AP/growth layer is shaped by tracer response, multipole response, AP convention, and spectrum-window handling. It is not a clean universal full-shape validation layer.

Diagnostic channel	Observed response	Interpretation
BAO coordinate wrapper	Quarter ruler improves the compressed BAO coordinate comparison	Strong coordinate-wrapper ruler diagnostic
Full sample spectrum shell	Locked quarter-down AP plus CPTG d_f improves the global shell	Positive but not universal; split models do better
BGS monopole $\ell = 0$	$q_{\rm iso}$ supports quarter-down in low-, mid-, high-, and all-k bins	Ruler channel remains supported
BGS quadrupole $\ell = 2$	$q_{\rm iso}$ wants inverse/up in low-, mid-, high-, and all-k bins	Quadrupole AP/projection/window sensitivity
ELG transverse channel	$q_\perp$ wants up broadly, especially through $\ell = 2$ and combined blocks	Broad transverse tracer-response sensitivity
LRG/QSO	Locked quarter-down direction remains supported in low-, mid-, and all-k bins	Stable monopole/ruler-supported sector
Gross window/nuisance structure	BGS/QSO share $q_{\rm iso}$-type structure, while ELG/LRG share $q_\parallel, q_\perp$-type structure	Gross HDF5 structure alone does not explain the split

Table 21: DESI full-shape spectrum-shell diagnostics. These are comparison-layer diagnostics only, not full-shape validation claims.

The local k-bin and multipole diagnostics show that the BGS behavior is not a high-k-only nonlinear or Fingers-of-God effect. The BGS monopole supports the CPTG quarter-ruler direction, while the BGS quadrupole flips in every tested k-range. The ELG transverse $q_\perp$ response is also broad rather than isolated to high k. Therefore the full-shape spectrum AP/growth layer remains an exploratory spectrum-shell diagnostic pending official nuisance-preserving AP, RSD, tracer-window, and covariance wiring. This does not weaken the compressed ShapeFit coordinate pass or the BAO/monopole quarter-ruler support.

The DESI status is:

$$\begin{gathered}\text{compressed ShapeFit coordinate vector: passed,}\\ \text{BAO quarter-ruler coordinate wrapper: supported,}\\ \text{full-shape spectrum AP/growth: exploratory spectrum-shell diagnostic.}\end{gathered} \tag{194}$$

14 Validation Status and Falsifiability

The locked CMB comparison layer supports the empirical statement:

$$\begin{gathered}\text{CPTG reaches control-level CMB recovery}\\ \text{across Plik, decisive NPIPE CamSpec,}\\ \text{and external SPT/ACT combined-stack comparisons,}\\ \text{with } A_{\rm lens} = 1.\end{gathered} \tag{195}$$

This statement is empirical and coordinate-specific. It means that the locked CPTG row passes the evaluated high-ℓ, low-ℓ, lensing, posterior-proxy, combined-stack profile, and combined-stack posterior comparisons without changing the gates and without applying lensing-amplitude inflation.

The cold-sector comparison transport

$$\mathcal{D}_c = 0.9735353159758136 \tag{196}$$

is treated as a derived CMB comparison-layer quantity and is not listed as an unresolved validation item.

The DESI compressed-coordinate result supports the statement:

$$\begin{gathered}\text{CPTG passes the DESI DR1 compressed ShapeFit}\\ \text{coordinate comparison using official HDF5 likelihood containers,}\\ \text{with the BAO quarter-ruler supported as a coordinate-wrapper diagnostic.}\end{gathered} \tag{197}$$

The DESI full-shape AP/growth sector supports the more limited statement:

$$\begin{gathered}\text{full-shape AP/growth comparisons are retained as}\\ \text{exploratory spectrum-shell diagnostics pending official}\\ \text{nuisance-preserving AP, RSD, tracer-window, covariance,}\\ \text{and convention wiring.}\end{gathered} \tag{198}$$

The CMB and DESI claims therefore sit at different validation levels. The CMB likelihood stack supports a locked cleaned comparison-layer recovery statement across the evaluated Planck-family, decisive PR4 CamSpec, and external ground high-ℓ/lensing comparisons. DESI supports a compressed ShapeFit coordinate-level pass plus a strong BAO quarter-ruler coordinate-wrapper diagnostic. The DESI full-shape AP/growth layer remains exploratory until the official nuisance-preserving full-shape likelihood machinery is wired consistently.

$$\begin{gathered}\text{CMB: locked cleaned comparison-layer pass through}\\ \text{high-}\ell\text{ profile robustness, Plik/SPT}\\ \text{posterior-proxy robustness, and combined-stack}\\ \text{profile/posterior validation,}\\ \text{DESI: compressed ShapeFit pass plus}\\ \text{BAO quarter-ruler coordinate-wrapper support.}\end{gathered} \tag{199}$$

The CMB branch would fail the present closure if the locked gates were held fixed and an independent likelihood stack of comparable constraining power gave

$$\Delta\chi^2_{\text{total}} > 10. \tag{200}$$

The branch would also be weakened if it required

$$A_{\text{lens}} > 1 \tag{201}$$

to recover high-ℓ or lensing structure. The present branch does not require this.

The DESI compressed-coordinate result would be weakened if the locked compressed vector failed under the official covariance containers. The DESI full-shape spectrum result is not treated as a failure of CPTG unless the same tracer- and multipole-dependent behavior remains after the official nuisance-preserving likelihood, tracer windows, AP conventions, redshift-space distortion sector, counterterms, stochastic terms, and covariance machinery are wired consistently.

Task	Required outcome	Status
Cold-sector comparison transport	Derive $\mathcal{D}_c$ from CPTG acoustic transport	Completed
Scalar-amplitude gate	Establish $\mathcal{A}_s = \pi/3$ as the scalar projection of the π-curvature branch	Geometric gate stated; numerical closure passed
Optical-depth gate	Establish $\mathcal{T}_\tau = \mathcal{G}_T^2$ as the two-step visibility contraction of the acoustic transport branch	Geometric gate stated; numerical closure passed
High-ℓ profile robustness	Preserve CMB recovery with fixed gates	Validation pass
Plik/SPT posterior-proxy robustness	Preserve recovery under MCMC sampling	Validation pass
Low-ℓ and lensing diagnostic stability	Preserve recovery without τ retuning or A_{lens} inflation	Diagnostic pass
Combined CMB stack profile validation	Preserve Plik, decisive CamSpec, and SPT/ACT recovery together	Validation pass
Gate-ablation robustness packaging	Demonstrate that the locked comparison gates are necessary and non-arbitrary under the tested comparison layer	Completed
Combined CMB stack posterior validation	Preserve combined-stack recovery under posterior sampling	Validation pass
Final CMB validation consolidation	Consolidate profile, posterior, convergence, and gate-closure evidence	Completed
DESI compressed coordinate closure	Preserve ShapeFit and BAO coordinate-wrapper success under locked conversions	Passed at coordinate level
DESI AP/growth spectrum-shell layer	Resolve BGS $\ell = 2$ and ELG $q_\perp$ response through official nuisance-preserving wiring	Exploratory
Raw DESI nuisance-preserving full-shape validation	Preserve tracer-bias, counterterm, stochastic, window, and covariance machinery	Open

Table 22: Validation status for the locked CPTG branch.

The CMB comparison requirements listed as passed or completed are satisfied without changing the locked gates, retuning τ, or using $A_{\text{lens}} > 1$. Within the locked comparison layer evaluated in this work, the CMB validation program is complete. The DESI full-shape AP/growth layer remains exploratory because the current spectrum-shell diagnostics are not a substitute for official nuisance-preserving full-shape likelihood evaluation.

15 Current Sector Status

The CPTG coordinate comparisons are summarized by sector-specific maps. Each sector compares a fixed CPTG object to the appropriate observational coordinate, likelihood summary, or compressed observable.

Sector	CPTG comparison object	Status
Background expansion	$E_\pi(a)^2$ and $E_{\mathrm{clock}}(a)^2$	Stable compressed branch
Age and clock map	Transported clock branch	Comparison supported
Supernovae	Standardized distance residuals with marginalized intercept	Full-covariance shape pass
BAO	D_M/r_d, D_H/r_d, D_V/r_d, AP coordinates, and $G_T^{-1/4}$ ruler conversion	Coordinate-wrapper comparison supported
BBN	Transport-scaled baryon and radiation conversion	Primary abundances within 2σ
Growth and lensing	$\mu(a)$, $\Sigma(a)$, $A_{\mathrm{bandpower}}$, and S_8 conversions	Compressed comparison supported
CMB	Locked cleaned acoustic-transport comparison row	Combined-stack profile/posterior pass across Plik, decisive NPIPE CamSpec, and SPT/ACT; exact gate-closure numerical pass
DESI ShapeFit and BAO	Compressed ShapeFit vector and BAO quarter-ruler coordinate wrapper	Coordinate-level pass and quarter-ruler support
DESI full-shape spectrum	AP/growth spectrum-shell response through tracer, multipole, and window layers	Exploratory diagnostic; official nuisance-preserving full-shape validation required

Table 23: CPTG comparison status by sector.

The theory remains falsifiable because each comparison layer has a fixed object and a defined failure condition. A CMB failure occurs if the locked CMB row becomes stressed with the gates fixed, or if recovery requires

$$A_{\mathrm{lens}} > 1. \tag{202}$$

A distance-sector failure occurs if the fixed distance relation misses covariance-level supernova or BAO structure after the correct comparison coordinates are applied. A BBN failure occurs if the transported baryon and radiation conversion misses primary abundance constraints outside the adopted uncertainty band. A growth or lensing failure occurs if the fixed perturbation and lensing reductions miss the compressed weak-lensing or CMB-lensing observables under their stated response maps.

A DESI compressed-coordinate failure occurs if the locked ShapeFit vector or BAO coordinate wrapper fails under the official covariance containers. A DESI full-shape failure requires the full AP, RSD, tracer-window, nuisance, counterterm, stochastic, and covariance machinery to be evaluated consistently. The spectrum-shell AP/growth diagnostics are therefore not promoted to raw full-shape validation by themselves.

The sector status can be summarized as

$$\begin{gathered}\text{CMB: locked cleaned comparison-layer pass through}\\ \text{combined-stack profile and posterior validation,}\\ \text{with Plik/SPT posterior-proxy robustness and gate closure,}\\ \text{DESI: compressed ShapeFit coordinate pass plus}\\ \text{BAO quarter-ruler coordinate-wrapper support,}\\ \text{full-shape AP/growth: exploratory spectrum-shell diagnostic.}\end{gathered} \tag{203}$$

16 Likelihood and Software Reference Layer

CMB and lensing comparison work requires a clear distinction between the fixed CPTG row and the software layer used to evaluate that row. The implementation references used for the likelihood and posterior machinery include the Cobaya external-likelihood interface [12], ACT DR6 lensing likelihood materials [13], ACT DR6 posterior-chain documentation [14], SPT likelihood software [15], and recent discussion of Planck likelihood choice in cosmological analyses [16]. These references describe evaluation infrastructure and likelihood context; they do not supply adjustable CPTG branch parameters.

17 Conclusion

The comparison-coordinate procedure is a disciplined translation from fixed CPTG-native quantities into the variables used by observational pipelines. The native branch is fixed first. Distances, acoustic rulers, CMB rows, BBN summaries, weak-lensing summaries, BAO coordinates, ShapeFit coordinates, and full-shape diagnostics are then expressed as coordinate-level tests of that branch.

The central comparison values are

$$p_C = \pi, \tag{204}$$

$$p_{\rm ac} = 3 - \frac{\pi}{100}, \tag{205}$$

$$\mathcal{G}_T = \frac{3 - \pi/100}{\pi}, \tag{206}$$

$$\mathcal{D}_c = 0.9735353159758136, \tag{207}$$

$$\mathcal{A}_s = \frac{\pi}{3}, \tag{208}$$

and

$$\mathcal{T}_\tau = \mathcal{G}_T^2. \tag{209}$$

A valid comparison preserves these fixed values unless a new branch is explicitly proposed and tested as a different theory object.

The locked CMB comparison layer now has control-level support from high-ℓ profile robustness, converged Plik/SPT posterior-proxy sampling, low-ℓ/lensing diagnostic stability, combined-stack profile validation, and combined-stack posterior validation across Plik, decisive NPIPE CamSpec, and SPT/ACT. These results hold with

$$A_{\text{lens}} = 1, \tag{210}$$

and without retuning τ, $\mathcal{D}_c$, $\mathcal{A}_s$, or $\mathcal{T}_\tau$. The exact gate-closure check also confirms the numerical map from $\omega_{c,\text{CPTG}}h^2$, $A_{s,\text{CPTG}}$, and τ_{CPTG} to the locked effective CMB row.

The non-CMB compressed sectors remain separated by comparison object. Pantheon+ gives a full-covariance distance-shape pass, BBN gives a primary-abundance pass within the adopted two-sigma comparison band, and compressed weak-lensing and CMB-lensing amplitudes remain within representative two-sigma comparisons. DESI gives a compressed ShapeFit coordinate-level pass and a strong BAO quarter-ruler coordinate-wrapper diagnostic. The DESI full-shape AP/growth sector remains an exploratory spectrum-shell diagnostic until the official nuisance-preserving AP, RSD, tracer-window, nuisance, counterterm, stochastic, and covariance machinery is evaluated consistently.

The status language is intentionally limited. A CMB comparison-layer validation pass, a Pantheon+ distance-shape pass, a BBN primary-abundance pass, a compressed weak-lensing diagnostic pass, a DESI ShapeFit coordinate-level pass, or a BAO quarter-ruler diagnostic pass means exactly what the corresponding coordinate test implements. It does not automatically imply a full raw-likelihood validation in every observational sector. That discipline keeps the comparison layer useful, falsifiable, and scientifically transparent.

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